MODULE 2 - Natural Resource Management
Advances in Energy Systems (Merits, Demerits, Global Status and Applications): Hydrogen, Solar, OTEC, Tidal and Wind. Natural Resource Management (Concept and case-studies): Disaster Management, Sustainable Mining - case studies and Carbon Trading

SESSION	Topics	BL	CO
01	Advances in Energy Systems - Hydrogen & Solar	**L2**	**CO2**
02	Advances in Energy Systems - OTEC, Tidal and Wind	**L2**	**CO2**
03	Natural Resource Management	**L2**	**CO2**
04	Disaster Management	**L2**	**CO2**
05	Sustainable Mining - case studies	**L2**	**CO2**
06	Carbon Trading	**L2**	**CO2**

Advances in Energy Systems

1) Hydrogen

Merits:
Clean Energy Carrier: Hydrogen is a clean energy carrier that produces only water vapor when used in fuel cells or combustion.

Versatility: Hydrogen can be used in various sectors, including transportation, industry, and power generation.

High Energy Density: Hydrogen has a high energy density, making it suitable for energy storage and transportation.

Demerits:
Production Challenges: Most hydrogen is currently produced using fossil fuels, leading to carbon emissions (grey hydrogen). Green hydrogen, produced using renewable energy, is still relatively expensive.

Storage and Transportation: Hydrogen has low energy density per unit volume, and it can be challenging to store and transport efficiently.

Global Status:
Production: Hydrogen production is increasing globally, with a focus on green hydrogen from renewable sources.

Applications: Used in industries, transportation (hydrogen fuel cell vehicles), and power generation.
Transportation: Fuel cell vehicles (cars, buses, trucks, trains).

Industry: Hydrogen as a feedstock for industries like steel and chemical production.

Power Generation: Electricity generation using fuel cells.

2) Solar

Merits:
Renewable: Solar energy is abundant, inexhaustible, and renewable.

Low Environmental Impact: Solar power generation has minimal environmental impact compared to fossil fuels.

Decentralized Generation: Solar panels can be installed on rooftops, enabling decentralized electricity generation.

Demerits:
Intermittency: Solar power generation is dependent on sunlight, resulting in intermittency and the need for energy storage solutions.

Land Use: Large-scale solar farms may require significant land use.

Global Status:
Capacity: Solar capacity is growing rapidly worldwide, with a significant increase in photovoltaic installations.

Cost Reduction: Solar panel costs have declined, making solar power more competitive.
Applications:

Residential: Rooftop solar panels for residential energy needs.
Commercial: Solar installations for businesses and industries.
Utilities: Large-scale solar farms for grid electricity generation.

3) OTEC (Ocean Thermal Energy Conversion):

Merits:
Renewable: OTEC harnesses the temperature difference between the ocean's surface and deep waters.

Stable and Predictable: Unlike solar & wind, OTEC provides a stable and predictable source of energy.

Demerits:
High Initial Costs: OTEC systems require substantial upfront investment.
Location Dependence: Suitable sites for OTEC are limited to tropical and subtropical regions.

Global Status:
Development: OTEC is in the early stages of development, with a few pilot projects in tropical regions.

Research: Ongoing research is focused on improving efficiency and reducing costs.
Applications:

Power Generation: Electricity production for islands and coastal regions.
Desalination: OTEC can facilitate desalination processes.

4) Tidal:

Merits:
Predictability: Tidal energy is predictable and less variable than solar and wind energy.
High Energy Density: Tidal currents have high energy density.

Demerits:
Environmental Impact: Tidal energy systems can have environmental impacts on marine ecosystems.
Limited Sites: Suitable sites for tidal power are limited to areas with strong tidal currents.

Global Status:
Pilot Projects: Tidal energy is in the early stages of commercial development, with pilot projects in several countries.
Research: Ongoing research focuses on improving technology and minimizing environmental impacts.

Applications:
Power Generation: Tidal stream and tidal range systems for electricity generation.

5) Wind:

Merits:
Renewable and Abundant: Wind is a renewable and abundant resource.
Mature Technology: Wind turbine technology is well-established and continually improving.

Job Creation: Wind energy projects contribute to job creation in manufacturing, installation, and maintenance.

Demerits:
Intermittency: Wind power generation is intermittent and variable.
Visual and Noise Impact: Wind turbines can have visual and noise impacts, particularly in densely populated areas.

Global Status:
Capacity: Wind power capacity has grown significantly globally, with both onshore and offshore installations.

Competitive Costs: Onshore wind is one of the most cost-competitive sources of electricity.
Applications:

Onshore Wind Farms: Wind turbines on land for electricity generation.
Offshore Wind Farms: Wind turbines installed in bodies of water for increased capacity and reduced visual impact.

In summary, each of these energy systems has its own set of merits and demerits, and their global status and applications vary. The transition to a more sustainable energy future often involves a mix of these technologies to address the challenges associated with intermittency, energy storage, and environmental impact. Ongoing research and technological advancements continue to shape the future of these energy systems

Sustainable Mining:
Concept: Sustainable Mining focuses on minimizing the environmental and social impacts of mining activities while ensuring economic viability. It involves responsible extraction, efficient resource use, ecosystem restoration, and social engagement.

Case Study: The Porgera Gold Mine, Papua New Guinea
Background: The Porgera Gold Mine faced environmental and social challenges, including deforestation, water pollution, and tensions with local communities.

Sustainable Mining Initiatives:

Community Engagement: The mine implemented programs to benefit local communities, such as infrastructure development, healthcare, and education.

Environmental Management: Initiatives included reforestation efforts, waste management systems, and measures to reduce water pollution.

Certification: Seeking and obtaining certification, such as ISO 14001, to demonstrate adherence to environmental management standards.

Carbon Trading:

Concept: Carbon Trading, or emissions trading, is a market-based approach to reduce greenhouse gas emissions. It involves the buying and selling of carbon credits, where entities with lower emissions can sell their excess allowances to those exceeding their allocated limits.

Case Study: The European Union Emissions Trading System (EU ETS)

Background: The EU ETS, established in 2005, is the world's first major carbon market. It covers various industries, including power generation, manufacturing, and aviation.

Carbon Trading Initiatives:
Cap and Trade: Industries are assigned emission allowances (caps), and those exceeding their limits must buy credits, while those below can sell their surplus.

Market Mechanism: The system encourages industries to invest in cleaner technologies and reduce emissions to stay within their allocated limits.

Expansion and Reforms: The EU ETS (European Union Emissions Trading System) has undergone expansions and reforms to improve effectiveness and align with evolving climate goals.

Natural Resource Management:

Concept: Natural Resource Management (NRM) involves the sustainable utilization, conservation, and protection of natural resources to meet the needs of present and future generations. It encompasses the responsible use of land, water, air, minerals, and other resources, considering ecological, economic, and social dimensions. Effective NRM aims to balance resource use with environmental conservation, promote sustainable practices, and mitigate negative impacts.

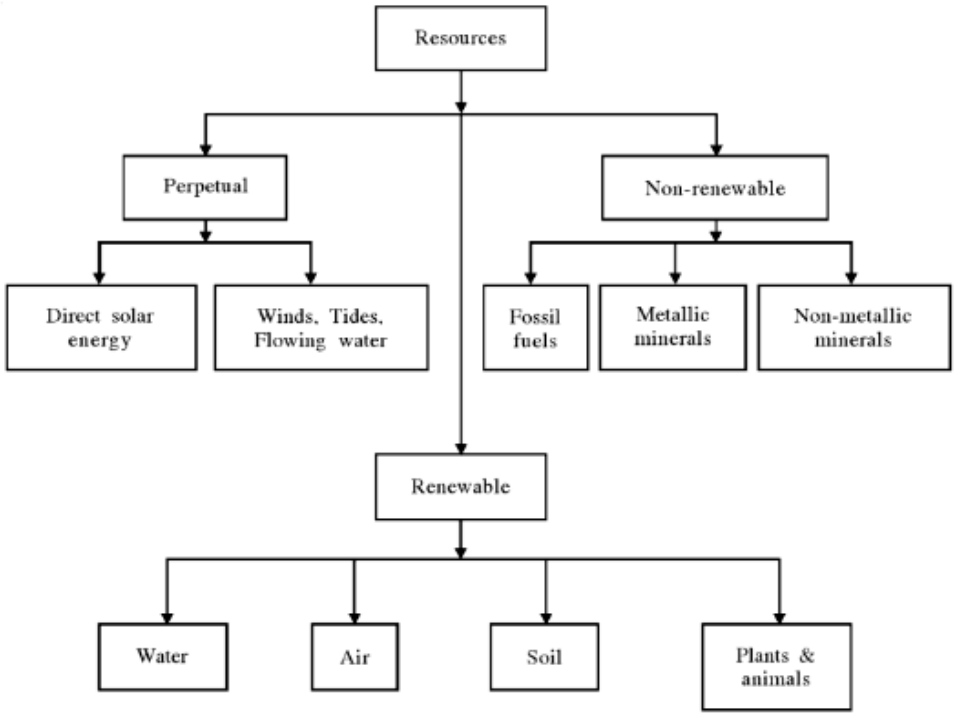
In summary, Natural Resource Management involves a holistic approach to balancing resource use and conservation. Disaster Management, Sustainable Mining, and Carbon Trading are key components within this framework, each addressing specific challenges through various initiatives and strategies. Case studies provide real-world examples of how these concepts are implemented and adapted to achieve sustainability goals.

Natural resources are the materials that are needed to sustain life and livelihood on Earth. They include air to breathe, water to drink, land to live on, walk on and grow food on, forest for timber, paper, wood products and medicinal plants and minerals for metals, fuel, energy etc.

Everything that human beings use is drawn from the natural resources. Hence, the human population is making an increasing demand over the natural resources. The increasing demand is attributed to the increased standards of living and excessive consumption of natural resources. Due to these factors the natural resources are depleting at a faster rate both quantitatively and qualitatively, which draws serious attention for conservation.

By the law of conservation, there is no resource that is spontaneously generated from nothing. Everything must come from somewhere and go somewhere. The only way by which the resources can be maintained indefinitely is through some form of recycling and reuse. The natural resources (Figure) fall into three general categories as:

- renewable,
- perpetual or non-renewable that can be recycled and
- non-renewable that cannot be recycled.



Disaster Management

Introduction

Natural calamities, of different types and intensities affect nations all over the world. Because of the large geographical size of the country, India often faces natural calamities like floods, cyclones and drought occurring fairly frequently in different parts of the country.

At times, the same area is subjected to floods and drought situation in successive seasons or years. While not all natural calamities can be predicted and prevented, a state of preparedness and ability to respond quickly to a natural calamity can considerably mitigate loss of life and property and human suffering, and restore normalcy at the earliest. It is, therefore, of paramount importance that a plan of action for dealing with contingencies that arise in the wake of natural calamities is formulated and periodically updated.

Types of Natural Calamities

Different natural, calamities can be distinguished from each other in terms of their nature and extent of their impact as described below.

Calamities like earthquakes, hailstorms, avalanches, landslides, tsunamies, etc. occur quite suddenly but they are restricted in their impact in terms of time and space.

Though floods and cyclones occur with some element of warning yet their occurrence is confined in duration.

Drought spans over a much longer time-frame and its adverse impact on the economic activities and life of an area is of a more lasting nature. The measures required to meet the threats posed by different calamities, therefore, differ considerably in terms of disaster preparedness and management

Major and Minor Calamities:

Natural calamities may be broadly grouped into major and minor types depending upon their potential to cause damage to human life and property. While natural calamities like earthquakes, droughts, floods, tsunamies and cyclones could be regarded as major, hailstorms, avalanches, landslides, fire accidents, etc. whose impact is localized and intensity of the damage is much less can be categorized as minor calamities.

Minor calamities like hailstorms, avalanches, landslides and fires also occur without any appreciable degree of forewarning and cause damage to properties and lives. However, areas prone to such disasters could also be identified and certain precautionary measures taken in the context of potential threat requiring general awareness and an ability to relate to a predefined system of appropriate responses on the part of the local administration.

Life Cycle Assessment (LCA)

It is a process of evaluating the effects that a product has on the environment over the entire period of its life, thereby increasing resource-use efficiency and decreasing liabilities.

LCA has its roots in the 1960s, when scientists concerned about the rapid depletion of fossil fuels developed it as an approach to understanding the impacts of energy consumption.

In the 1970s, the U.S. Environmental Protection Agency refined this methodology. At present, the ISO 14040 and 14044 standards describe the principles and guidelines for LCA.

LCA can be used to study the environmental impact of either a product or the function the product is designed to perform. LCA is commonly referred to as a "cradle-to-grave" analysis.

Thus, LCA studies the environmental aspects and potential impacts throughout the product's life, from raw materials acquisition through production, use and disposal.

The key elements of LCA are the following:

a) Compiling an inventory of relevant inputs and outputs of a product system;
b) Evaluating the potential environmental impacts associated with those inputs and outputs;
c) Interpreting the results of the inventory analysis and impact assessment phases in relation to the objectives of the study.

MINING

Mining is the act of extracting ores, coal, etc. from the earth. Mining on an industrial scale can produce environmental damages resulting from exploration and development, even long after the mine is closed.

The following is a list of adverse impacts of mining on forests and the environment.

- Mining, especially open-pit mining, generates enormous quantities of waste compared to any other natural resource extraction activity. Water interacts with these wastes to generate contaminated fluids that can pollute soils, rivers, and ground water.
- Erosion and sedimentation present another environmental issue for mine sites.
- Increased access to otherwise remote forest areas and provide an opportunity for further activities, especially in places where population pressures already exist.
- Dust generated from mining activities can cause air pollution, a serious cause of illnesses, generally in the form of respiratory troubles in people and asphyxia of plants and trees. Furthermore, usually, release of gases and toxic vapour takes place.
- Mining activities consume enormous quantities of timber for their construction and as a source of energy. Mining activities imply major works such as road building, ports, mining villages, deviation of rivers, construction of dams and energy generating plants.
- The deafening sound of the machinery used in mining and the blasting create conditions that may become unbearable for the local population and the forest wildlife.

- The large disturbances caused by mining can disrupt environments, adversely affecting aquatic habitats (i.e. lakes, ponds, streams, rivers), terrestrial habitats (i.e. deserts, grasslands, forests), and wetlands that many organisms rely on for survival.

- As large-scale mining creates new infrastructure and provides additional employment, permanent settlements can arise around these operations in areas that otherwise would have remained more sparsely inhabited. Mining comes along with its promise of wealth and jobs, but millions of people throughout the whole world testify to the high social costs that it brings with it.

The following is a list of the negative social impacts of mining.

- Appropriation of the land belonging to the local communities
- Impacts on health
- Alteration of social relationships
- Destruction of forms of community subsistence and life
- Social disintegration ∑ Radical and abrupt changes in regional cultures
- Displacement of other present and/or future local economic activities

Multiple Choice Questions (MCQ)

1) Who heads the National Crisis Management Committee?
a) Prime Minister b) President **c) Cabinet Secretary** d) Ministry of Environment

2) When did the Disaster Management formed?
a) 2005 b) 2006 c) 2007 d) 2008

3) Where is International Tsunami Center located?
a) New York b) London c) Mysore **d) Honolulu**

4) 21. South Africa is leading exporter of which mineral?
A) Copper B) Diamond C) Silver **D) Gold**

5) Which option is correct, when we only accomplish two out of three pillars of Sustainable Development?
A) Economic + Environmental Sustainability = Viable
B) Social + Environmental Sustainability = Bearable
C) Social + Economic Sustainability = Equitable
D) All of the above

6) The word 'Sustainable Development' came into existence in the year?
A) 1992 B) 1978 **C) 1980** D) 1987

7) 24. The United Nation's Commission on Sustainable Development (CSD) was started by the UN General Assembly in the year.
A) 1995 B) 1994 C) 1993 **D) 1992**

8) 25. After mining the huge holes left behind is used for
A) Wastewater storage B) Waste and water storage
C) Waste Disposal D) Waste Storage

9) 26. Mercury and lead are toxic elements, which causes
A) Noise Pollution B) Air Pollution
C) Water Pollution **D) Land contamination**

10) 27) The other word of Landscaping is
A) a) Reduction b) **Restoration** c) Removing top soil d) None

11) Whenever the price of the mineral remains high, the firm seeks for
a) New countries b) Remains the same c) New miners **d) New deposits**

12) 29) Which of the option is not incorporated as sustainable development parameters?
a) Gender disparity and diversity b) Inter and Intra-generation equity
c) Carrying capacity **d) None of the above**

13) 30) Cloud seeding with silver iodide is based on the
a) **Bergeron process** b) collision-coalescence process
c) both a and b d) none of these

14) Why do clouds represent a source of heat for the atmosphere?
a) They increase the absorption of solar radiation
b) They conduct heat from the earth's surface
c) **Cloud formation releases latent heat to the air**
d) Melting ice crystals in the cloud absorb heat

15) What is cloud seeding?
I. The addition of substances to the cloud to make rain
II. The addition of substances to the cloud to make silver
III. The taking away of substances from clouds to make rain

a, III b. I, II c. III only d. II only **e. I only**

16) Which of the following is used to seed warm clouds?
a. Platinum b. Gold **c. Silver iodide** d. Dry ice e. Salt

17) Which of the following has been used to seed clouds?
i. silver iodide ii. Dry ice iii. Salt

1. ii only **2. i only** 3. i & ii only 4. all of the above

18) The cloud droplets in a cloud are formed by water vapor molecules and
a) molecules of air b) other water vapor molecules
c) hygroscopoic nuclei d) protons e) ions

19) How much of India's coastline is vulnerable to disasters?
a) 5000 km b) 5500 km **c) 5700 km** d) 6000 km

20) How much percentage of the landmass is prone to earthquakes in India?
a) Around 58% b) Around 60% c) Around 63% d) Around 65%

21) How many hectares of Indian land are prone to floods and river erosion?
a) Around 25 million hectares b) Around 30 million hectares
c) Around 35 million hectares **d) Around 40 million hectares**

22) When did National Disaster Management Authority formed?
a) 2000 **b) 2005** c) 2010 d) 2015

23) Hazards and Disasters are mainly classified as
a. Physical and chemical b. **Natural and Human induced**
c. Physical and Human d. Social and cultural

24) A hazard is a situation where there is
a. Threat of natural calamity **b. Threat to property and lives from calamities**
c. Threat for consequences of disaster d. All of the above

25) Disaster is an event arising out of
a. **Result of hazard event** b. Causes of hazard event
c. Causes of disaster event d. All of the above

26) The typical examples of man-made disasters are
a. Chemical explosion b. Leakage of toxic waste
c. War and civil strife **d. All of the above**

27) What are the consequences of disaster on a society?
a. Loss of life b. Damage to property
c. Environmental Damages **d. All of the above**

28) The scientific study of earthquake is called
a. Seismograph **b. Seismology** c. Both a and b d. None of the above

29) Disaster Management includes:
a. Mitigation b. Reconstruction c. Rehabilitation **d. All of the above**

30) The term 'disaster' is derived from which of the following language?
a. Greek b. Latin **c. French** d. Arabic

31) Disaster Management covers?
a. Maintaining control over disasters
b. Reducing the effects of disasters
c. Briefing the top officials of govt. on the effect of disasters
d. All the above

32) Mass killing diseases can be referred as
a. Biological disaster b. Industrial disaster
c. War disaster d. Flood disaster

33) What are the important measures to be taken in community level of disaster preparedness
a. Increased awareness b. Provision of early and timely warning
c. Landuse planning **d. All of the above**

34) Under which Ministry National Disaster Management Authority comes?
a) Ministry of Environment b) Ministry of Foreign Affairs
c) Ministry of Pollution **d) Ministry of Home Affairs**

35) Disaster Management refers to manage disaster response in the country.
a) True b) False